

the new elements of `<audio>` and `<video>`. It also contains an optional `<legend>` element performing the function of a caption. Our semantic class name version could be like what follows:

```
<div class="figure">
  
  <p class="legend">[...]</p>
</div>
```

Extensible semantics

Many pages around the Web use micro-formats to add more structured semantics than what's available in HTML's impoverished set of elements and attributes. In this case, the values used for the class attribute come from agreed-upon vocabularies, sometimes adopted from other standards, such as vCard, and sometimes from newly minted vocabularies where no solid pre-existing standard exists.

This is a real problem that needs to be solved here. We need mechanisms in HTML that clearly and unambiguously enable developers to add richer, more meaningful semantics—to their mark-up. This is perhaps the single most pressing goal for the HTML 5 project.

But it's not as simple as coming up with a mechanism to create richer semantics in HTML content: there are significant constraints on any solution. Perhaps the biggest one is backward compatibility. The solution can't break the millions of browsing devices in use today, which will continue to be used for years to come. Any solution that isn't backward compatible won't be widely adopted by developers for fear of excluding readers.

Canvas—inline SVG and MathML

The `<canvas>` element is another exciting addition to the HTML 5 specifications, second only to `<video>` (which will be discussed later in the article). HTML 5 Canvas gives you an easy and powerful way to draw using JavaScript. For each canvas element

you can use a 'context' (think about a page on a drawing pad), into which you can issue JavaScript commands to draw anything you want. Browsers can implement multiple canvas contexts and the different APIs provide the drawing functionality.

Most of the major browsers include the 2D canvas context capabilities—Opera, Firefox, Konqueror and Safari. In addition, there are experimental builds of Opera that include support for a 3D canvas context.

The baby steps

Creating a canvas context on your page is as simple as adding the `<canvas>` element to your HTML document. Here's an example:

```
<canvas id="CanvasID" height="300"
width="900">
```

Show this text if the browser is not Canvas compatible
 </canvas>

It is advisable to define an element ID. This will be helpful later in referencing the element in the JS code. Moreover, the height and width of the canvas also needs to be defined.

Now that you have managed to create your very first canvas, let's shake things up a bit. To draw inside your canvas, you need to use JavaScript. First find your canvas element using `getElementById`, then initialise the context you want. Once you do that, you can start drawing into the canvas using the available commands in the context API. Let us now learn to draw a little rectangle inside the canvas we just created.

```
// Referencing the 'CanvasID' element
var elem = document.getElementById('CanvasID');
```

```
// Checking for browsers
if (elem && elem.getContext){
  // You can only initialize one context per element.
  var context = elem.getContext('2d');
  if (context){
    // Here comes the rectangle, now. Remember
```

p00r JOB MARKET ?
 Campus Placement ?

Future is OPEN

Special Offer
 for 2009 passed out
 Placement assistance*



RHCE

~~Rs 14,500/-~~ **Rs 10,500/-**
 Free Training for RHCE Exam
 RHCE Exam : 13 & 28 Aug 09

LAMP Pro
 (Linux+Apache+MySQL+ PHP)
~~Rs 12,500/-~~ **Rs 10,500/-**



Authorized Training Partner

Zend Certification

ZCE - International Certification for PHP
 from Zend Technologies - USA

Save

Up to
45% Now

~~Rs 45,000/-~~ **Rs 24,500/-**

Hostel Facility available
 For our station students



Lynus Academy Pvt Ltd
 1961-B, Vijaya Complex - 5th Floor
 Asiatic Colony, Anna Nagar - West
 Chennai 600101

044-42171278
9952966527

contactus@lynusacademy.com
 www.lynusacademy.com

* Conditions Apply